

Quiz: Instruction formats

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Started on	Tuesday, January 14, 2025, 11:46 PM
State	Finished
Completed on	Tuesday, January 14, 2025, 11:59 PM
Time taken	12 mins 58 secs
Grade	1.00 out of 5.00 (20%)

Question 1

Incorrect

0.00 points out of 1.00

In an SI-format instruction

Select one:

- ☐ A. One operand is a register and another a 16-bit relative address
- ☐ B. Both operands are registers
- ☐ C. One operand is a storage address and another an immediate value
- ☐ D. One operand is a register and another an indexed storage address
- ☐ E. Both operands are storage addresses
- ☒ F. One operand is a register and another a base-displacement storage address 
- ☐ G. One operand is a register and another a long displacement storage address

The correct answer is: One operand is a storage address and another an immediate value

Question 2

Incorrect

0.00 points out of 1.00

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The SS instruction 'MVC 20(10,3),8(12)', will be, in memory (recall that the MVC opcode is X'D2')

Select one:

- ☐ A. D202 3014 C008
- ☐ B. D20B 3014 C008
- ☒ C. D210 3014 C008 ✖
- ☐ D. D209 3014 C008
- ☐ E. D2AA 3014 C008
- ☐ F. D20A 3014 C008

All the options have:

- The same opcode (X'D2')
- The same first operand addressing halfword (X'3014'), meaning base register 3, displacement 20.
- The same second operand addressing halfword (X'C008'), meaning base register 12, displacement 8.

So, we have to choose the option with the correct operand length (the byte following the opcode).

In SS-type instructions, the operand length is the actual length minus one (X'00' specifies a 1-byte length; X'FF' specifies 256 bytes).

In this case, the length is 10 (decimal), so the correct option is the one that specifies X'09' as the operand length.

The correct answer is: D209 3014 C008

Question 3

Incorrect

0.00 points out of 1.00

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In RI-type instructions, the second operand is a long displacement 32-bit fullword

Select one:

- ☒ True ✕
- ☐ False

RI-type instructions use neither long displacement nor base-displacement. They have **immediate** operands.

In the examples we've seen, those immediate operands are **relative addresses**.

The correct answer is 'False'.

Question **4**

Incorrect

0.00 points out of 1.00

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In an RR-format instruction

Select one:

- ☐ A. Both operands are registers
- ☒ B. One operand is a register and another a long displacement storage address ✘
- ☐ C. Both operands are storage addresses
- ☐ D. One operand is a register and another a 16-bit relative address
- ☐ E. One operand is a register and another an indexed storage address
- ☐ F. One operand is a register and another a base-displacement storage address
- ☐ G. One operand is a storage address and another an immediate value

The correct answer is: Both operands are registers

Question **5**

Correct

1.00 points out of 1.00

In z/Architecture, the first byte of an instruction always contains the Opcode (Operation Code)

Select one:

- ☒ True ✓
- ☐ False

Correct!

The correct answer is 'True'.